



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2025	M.TECH Dual Degree 5Y	IIT Kharagpur	8.27 / 10
2019	CBSE Class XII (AISSE)	St.Antony's Public School	95.4%
2017	CBSE Class X (AISSE)	Rajagiri Christu Jayanthi Public School	10 / 10

INTERNSHIPS

Generative AI Intern | Perceptive Analytics

Jun '24 - Jul '24

Objective: Developed an AI-driven SaaS for efficient document processing to accelerate the motor insurance claims processing pipeline

- Created a **Retrieval-Augmented Generation** system, integrating the **FAISS** vector database and **OpenAIEmbeddings** for data retrieval
- Achieved a **30%** cost reduction on **OpenAI API** calls using the **GPT- 4o mini** model with comparable results to standard GPT-4 models
- Utilized **LangChain** for document parsing, chunking, data retrieval and deployed on **Streamlit**, offering an interactive interface for users
- Improved retrieval accuracy by using **MultiQueryRetrieval** to generate 5 more queries mitigating the limitation of distance-based retrieval.

Computer Vision Intern | Parimitra | Indian Institute of Technology (IIT), Roorkee

Jun '24 - Jul '24

Objective: Developed an edge computing device for smart, real-time crowd monitoring and dynamic alert system to enhance public safety

- Achieved a **0.92 Mean Average Precision (mAP)** by training the small **YOLOv8** model on a novel Indian crowd dataset of **35000** images
- Deployed the quantized **TensorFlow Lite** model on Raspberry Pi, integrating the **ByteTracker** algorithm for multi-object tracking (MOT)
- Reduced the inference time from **1000 ms** to **100 ms** by integrating **Google Coral TPU**, to optimize the device's overall performance
- Designed a data management schema using **Google Cloud Services** to store crowd analytics data and device diagnostics of **8 devices**

Computer Vision Intern | National Institute of Technology (NIT) Trichy

Jun '23 - Jul '23

Objective: Developed an image denoising model to improve the clarity of plant leaf images, enabling early and accurate disease detection

- Developed an **image denoising** model based on the **U-Net** architecture, utilizing **CNN** and **Upsampling layers** with skip connections
- Engineered a custom dataset using **scikit-image** library to generate noise patterns like **Gaussian** and **Salt-and-Pepper** onto clean images
- Achieved a **Peak Signal-to-Noise Ratio (PSNR)** of **31.38** and a **Structural Similarity Index (SSIM)** of **0.89** showing notable image quality

PUBLICATIONS

Image Super Resolution | CERA '23, IEEE International Conference

Jun '23 - Jul '23

S. Deivalakshmi, J. George. et.al.(2023) "Image super resolution using Enhanced Attention Block (EAB) for Grape Leaf Disease Detection,"

- Developed a model architecture of a **Multistage Hierarchical Feature Encoding** module which includes **attention** and **residual** learning
- Attained high classification accuracy of **99.58%** and **98.71%** for downsampling factors of 2 and 4 surpassing ResNet50 and GoogleNet

COMPETITION/CONFERENCE

Dr, APJ Abdul Kalam Young Research Fellowship 2024 | Prof. Mahesh Mohan

Jan '24 - Apr '24

Qualified for the finals being among the top 10 researchers across India and presented the research work before an esteemed jury panel

- Translated a code that implements a **Complex-Valued Neural Network(CVNN)** for **Synthetic Aperture Radar(SAR)** data reconstruction
- Evaluated a **coherency mean** of **0.92** versus 0.0246 demonstrating the impact of image **reconstruction** loss in the classifications task
- Evaluated a **5%** increase in classification accuracy for **multi-channel(HH and HV)** polarization as compared to single-channel polarization

PROJECTS

Semi Markov Model for Change Point Detection in Time Series | Course Project

Jul '24 - Present

Objective: Explored statistical modeling methods to identify market trends, conditions to improve trading outcomes and drive higher returns

- Implemented a three-state **Hidden Semi-Markov Model (HSMM)** to decode different market conditions into bear, bull, &sidewalk states
- Developed a **trading strategy** using outputs from HSMM model to systematically generate trading signals across market conditions
- Calculated the duration a stock spends in each state to assess market behaviour and price trends, aiding in **long-short** trading strategy

Non-Speech Emotion Recognition | University of Tartu, Estonia | Prof. Rajesh Sharma

Jan '24 - Apr '24

Objective: Coducted a study to explore the effectiveness of audio encoders from multi-modal models for non-verbal emotion recognition

- Assessed the quality of extracting audio embeddings using **x-vector**, **WavLM**, **Wav2Vec**, and **UniSpeech** deep neural audio encoders
- Implemented **SVM**, **Random Forest**, and a **CNN** model to evaluate the extracted embeddings in **non-verbal emotion recognition** tasks

SKILLS AND EXPERTISE

Programming Languages : Python | SQL | C | C++ **Tools :** Google Firebase | MySQL | Github | SSH | AWS | Sagemaker | Microsoft 365**Python Libraries and Frameworks :** NumPy | Pandas | SciPy | Matplotlib | Seaborn | PyTorch | Tensorflow | Keras | OpenCV | Scikit - Learn | XGBoost | Streamlit | Flask | FastAPI | NLTK | SpaCy | Ultralytics | Transformers | LangChain | LlamaIndex | Beautiful Soup | DVC

AWARDS AND ACHIEVEMENTS

- Secured an **All-India Rank of 8327** out of over 2,00,000+ qualified candidates in the JEE Advanced 2020 Exam, conducted by IIT Delhi.
- Secured a **State Rank of 35**(Open Category) in CEE Kerala Engineering & Pharmacy (KEAM) Exam conducted by the Government of Kerala.

EXTRA CURRICULAR ACTIVITIES

- Captain of the **Silver-winning** choreography team, representing JCB Hall of Residence, among **12** teams in the General Championship '23
- Actively engaged as a theatre practitioner with the English Drama Society and a performing artist with the Dance Society at IIT Kharagpur
- Part of the **Gold-winning** choreography team, representing IIT Kharagpur, in **Centrifuge Spring Fest '23** an All-India Choreography event
- Part of the **Silver-winning** football team, representing JCB Hall of Residence, in the General Championship'23 competing against **12** teams